Haussohou Daniel Edi

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Professional Summary

Psychometrician with experience supporting large-scale educational and high-stakes professional certification examinations. Expertise in Rasch measurement, Many-Facet Rasch, IRT, CTT, item calibration, scoring, equating, scaling, DIF, reliability, and validity studies. Advanced SAS and R programmer with experience conducting operational psychometric analyses, quality control, technical reporting, and research studies. Co-founder of Kheenoukh Education, a curriculum-aligned assessment program for middle-school mathematics in Côte d'Ivoire. Proven ability to collaborate with psychometricians, subject matter experts, engineering, product, and test development teams to deliver accurate, high-quality assessment solutions.

Technical Skills

- Psychometrics: Rasch Measurement, Many-Facet Rasch, IRT, CTT, CAT, MST, Equating, Scaling, Vertical Scaling, DIF, Reliability, Validity
- Programming: SAS, R
- Software: IRTPRO, Winsteps, FACETS, STUIRT, POLYCSEM
- Assessment: Item Banking, Test Blueprints, Test Construction, Calibration, Scoring, Score Reporting, Quality Control, Technical Reporting
- Open-Source Software: Author and maintainer of six psychometric R packages on CRAN (decisionfacets, retestR, coldstart, driftwatch, throughyear, transDIF); version 0.2.0 added oral-exam designs with 0–100 scores and linking-sensitivity reports; portfolio at edidatasolutions.github.io
- Languages: French (native), English (professional fluency)

Experience

Psychometrician, National Medical Specialty Certification Board *10/2026–Present*

- Conduct psychometric analyses supporting the development, administration, scoring, and reporting of the board's certification examinations, including item analysis, equating, standard-setting support, and post-administration review.
- Perform quality assurance reviews, data validation, and technical documentation for examination, survey, and research datasets.
- Prepare tables, visualizations, and written summaries for committee discussion, Board reporting, and operational decision-making.
- Support item writer, oral examiner, and committee activities through data analysis and meeting preparation, and collaborate with data staff and vendors on data exchanges and operational workflows.
- Contribute to survey-based and examination research and to the automation of reproducible analytic workflows.

Co-founder, Kheenoukh Education (Côte d'Ivoire) *2025–Present*

- Built a bank of 1,432 multiple-choice items in middle-school mathematics (grades 6–9), each reviewed against the official Ivorian curriculum and documented with DOK level, assessed skill, rationale, and a-priori Rasch difficulty.
- Designed the blueprint of 43 test forms (3 diagnostic, 36 formative, 4 summative) linked by vertical anchor items, a provisional Rasch-based vertical scale (160–350), and four performance levels with provisional cut scores.
- Led the design and deployment of a mobile web platform that delivers the assessments and produces automated student, parent, and school score reports, in preparation for a free 2027–2028 pilot in Abidjan.

Psychometrician, North Carolina Department of Public Instruction (NCDPI) *09/2024–09/2026*

- Led operational psychometric analyses for statewide EOG/EOC ELA and mathematics assessments, achieving 100% completion of score verifications, item analyses, and technical reports across all cycles; rated Exceeds Expectations for two consecutive review periods.
- Conducted IRT and CTT item calibration using IRTPRO and Winsteps, along with scaling, equating, DIF analyses, reliability, and validity studies to support defensible statewide assessment results.
- Optimized SAS programs for item analysis and form-building, improving efficiency department-wide, and became the team's primary internal resource for SAS programming support.
- Collaborated with UNC-Greensboro's psychometric research team (led by Dr. Kyung Yong Kim) on quality control review of domain scoring; successfully transitioned the Domain Scores analysis in-house, replicating prior results and preparing the codebase for future administrations.

- Conducted retest and item drift research that directly informed state testing policy changes; presented findings at NCDPI Technical Advisory Committee meetings and in statewide webinars hosted by the Superintendent's office on test security and policy.
- Built and delivered new interim and summative test forms for EOG/EOC ELA and mathematics ahead of established deadlines, streamlining downstream test development and content review workflows.

Psychometrician (Contract), National Medical Specialty Certification Board *04/2026–06/2026*

- Supported the Principal Psychometrician in scoring the March and May 2026 Certifying Examinations: equating, item analysis, quality control, and summaries for the Board of Directors.
- Co-authored with the Principal Psychometrician an NCME 2027 proposal, "Evaluating Connectivity Strategies in Many-Facet Rasch Models."

Associate Research Scientist, Pearson *12/2022–09/2024*

- Conducted IRT and CTT item calibration (IRTPRO, Winsteps) and item bank maintenance for statewide programs.
- Conducted equating, linking, scaling, DIF, reliability, validity, and simulation studies, with supporting technical documentation.

Psychometric & Research Intern, Edmentum *06/2022–12/2022*

- Conducted item analyses, calibration, and form construction with content and product teams.

Psychometric & Research Intern, National Board of Chiropractic Examiners (NBCE) *06/2021–08/2021*

- Executed OSCE and written exam scoring, including rater effects analysis, and supported practice analysis studies.

Education

Ph.D., Applied Statistics & Research Methods – University of Northern Colorado

Dissertation: The Impact of Rater Effects on Decision Consistency and Decision Accuracy under the Many-Facet Rasch Measurement Framework.

M.S., Statistics – University of New Mexico

B.S., Petroleum Engineering (Minor: Mathematics) – University of Louisiana at Lafayette

Selected Research Presentations

- Asare, E. & Edi, D. Impact of Item Drift on Pre-Equated Scores: A Rasch-Based Investigation (NCME 2026)
- Asare, E., & Edi, D. (2025). School Characteristics and Socio-Economic Status in States with High- vs Low-Performing Students (NCME 2025)
- Edi, D., & Himelfarb, I. Impact of Rater Effects on Examinee Ability in Objective Structured Clinical Examinations. International Meeting of the Psychometric Society (IMPS 2023).
- Edi, D. & Powers, S. Impact of Item Parameter Drift on Ability Estimates in Computerized Adaptive Testing (NCME 2023)
- Edi, D. (2023). Impact of Rater Effects on Classification Consistency and Accuracy in Performance-Based Assessments (NCME 2023).

Submitted Proposals (NCME 2027, Under Review)

- Edi, D., & Joldersma, K. Evaluating Connectivity Strategies in Many-Facet Rasch Models.
- Edi, D. When Anchoring Both Facets Isn't Safe: A Certifying-Exam Equating Failure.
- Edi, D. Real Recovery or Composition Shift? Decomposing the Pandemic NAEP Trend.
- Edi, D. Resolving Simpson's Paradox in Accountability Ratings: A Simulation Study.

Professional Service

- Proposal Reviewer, National Council on Measurement in Education (NCME), 2025–Present
- Founder, Agora Scientifique (agorascientifique.org), a free online platform for Ivorian researchers and students